AI ACADEMY · LEVEL 5 · GRADE 10 · WEEK 12 · STUDENT LESSON

Training a small network

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Image-pattern Network Investigation

Before you start

Why should all parameters in one simultaneous update use the same saved gradient?

Plan time to read, investigate and explain. Keep predictions separate from observations.

STUDENT LESSON · 1 OF 18

Your mission

> Start with the target: Trace repeated forward loss gradient and update steps. Try the recall before opening notes.

Your learning target

I can trace repeated forward loss gradient and update steps.

I can keep **frozen parameter**s distinct from the trained parameter.

I can report **training** results with their evaluation limits.

Remember before reading

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from Updating parameters. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Slopes and local change. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Forward passes by hand. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

This week you will practise:

Trace repeated forward loss gradient and update steps

Keep **frozen parameter**s distinct from the trained parameter

Report **training** results with their evaluation limits

Retrieve what you already know

> Why should all parameters in one simultaneous update use the same saved gradient?

> **My starting explanation**

This week's end goal

By the end, I can explain **training** a small network using a traced example and a stated limitation.

Notice the evidence

> Investigate the supplied Training a small network case. Separate observations, given facts and assumptions before choosing an explanation.

Use one hidden ReLU unit h = ReLU(x) and an **output** y = vh. The hidden **input** weight is fixed at 1, its bias is fixed at 0, and the **output** bias is fixed at 0. Only **output** weight v is trained.

This is a restricted two-layer network, not **training** every parameter of a general network.

Training pairs are (x, target) = (1, 2) and (2, 4). Both x values are positive, so hidden activations are 1 and 2. Define ordinary MSE over both rows.

Start at v = 0 and use learning rate 0.1 for exactly four full-data updates. The declared budget prevents stopping only when a preferred result appears.

Update count	v	Training MSE
0	0	10
1	1	2.5
2	1.5	0.625
3	1.75	0.15625
4	1.875	0.0390625

Each row evaluates the parameter value after that many updates. Row zero records the starting model. The loss decreases in this specific experiment; that pattern is not a promise for all networks or rates.

> **List the given facts and one fact that is not supplied.**

See how it works

> Follow the mechanism for Training a small network. Connect each labelled input, deciding step and result.

Let errors be v - 2 and 2v - 4. MSE is their squared sum divided by 2, which simplifies to 2.5 times (v - 2) squared. Its gradient with respect to v is 5 times (v - 2).

At v = 0 the gradient is -10, so the first update gives v = 0 - 0.1 times -10 = 1.

```
train = [(1.0, 2.0), (2.0, 4.0)]
v = 0.0
rate = 0.1
for step in range(5):
    hs = [max(0.0, x) for x, target in train]
    errors = [v * h - target
              for h, (x, target) in zip(hs, train)]
    loss = sum(e * e for e in errors) / len(train)
    print(step, v, loss)
    if step == 4:
        break
    gradient = 2 * sum(e * h for e, h in zip(errors, hs)) / len(train)
    v = v - rate * gradient
```

The factor 2 comes from squared error; division by the number of rows makes this the gradient of the mean, not the sum. The code uses every **training** row for each update. It prints five states but performs only four updates.

All data are supplied fictional numbers.

Connect the picture and the explanation

Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

Compare the cases

> Compare the supplied cases for Training a small network. Identify the detail that changes the conclusion.

Compare a familiar case with a changed case

At v = 1, the sum of squared errors is 5. A learner uses gradient -10 with rate 0.1. Explain the mismatch with this lesson and give the next v under MSE.

Keep the procedure fixed. Explain what changed in the inputs and result.

Change one thing in the pictured case

Change one thing

Change: Change target from 3 to6 and restart v = 0 with the same rate.

Predicted result

Prediction: First gradient=-36; v becomes2; **prediction** 6 and loss 0.

Why the result follows

Reason: The hidden value remains3, but the target residual and required **output** weight double.

STUDENT LESSON · 5 OF 18

Words that unlock the idea

> Use the vocabulary of Training a small network to explain a concrete decision, rather than only memorising definitions.

Word	Meaning
Training loop	A repeated sequence of prediction, loss calculation, gradient calculation and parameter update
Frozen parameter	A model value intentionally held fixed during fitting
Training budget	A declared limit on updates or other training resources
Learning curve	A record of a performance measure across training steps or time

Use two terms to explain a step in this week’s example.

STUDENT LESSON · 6 OF 18

Follow a worked example

> Follow the worked trace for Training a small network, predict a step, then compare with its explanation.

One full-data update from v is 1

STAGE	COMPUTE WITH CURRENT V	RESULT
Forward and loss	Predictions [1, 2] for targets [2, 4]	Errors [-1, -2], MSE 2.5
Gradient	2 times (-1 times 1 - 2 times 2) / 2	Gradient -5
Update with rate 0.1	1 - 0.1 times -5	Next v is 1.5

Each row depends on the preceding calculation. Recompute predictions and loss on the next iteration; do not keep using the first gradient after parameters change.

> **Starting from v = 1.5, calculate both training predictions, errors, MSE, gradient and the next v.**

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Training a small network

Train a single **output** weight while clearly marking the frozen hidden computation.

The facts we will use

> **Supplied facts**
>
> Toy h=ReLU(x), y=vh. One **training** pair x = 3, target 3; start v = 0; squared loss; rate 1/18.

> **Supplied facts**
>
> All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

AI ACADEMY / AI-5/12

Training a small network

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Freeze hidden step

h=3 for this pair; only v is trainable.



2. Compute initial gradient

Loss9; gradient 2(3v-3) x 3=-18 at v=0.



3. Update and verify

v becomes1; prediction 3 and loss 0. Next gradient is 0.

CHANGE ONE THING

Change target from 3 to6 and restart v = 0 with the same rate.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Freeze hidden step	h=3 for this pair; only v is trainable.
2. Compute initial gradient	Loss9; gradient 2(3v-3)×3=-18 at v=0.
3. Update and verify	v becomes1; prediction 3 and loss 0. Next gradient is 0.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Why the result follows

Only the **output** weight v is being fitted in this toy network; the hidden ReLU is fixed. The **training input** gives hidden value three.

Squared-error slope points v from zero towards one, and this specially chosen rate reaches one in a single update. Zero loss on one **training** pair is exact **fitting** of that pair, not **evidence** about other **input**s.

What this example does not establish

Keep the stated limit

A zero loss on one example is memorised fit to that example, not proof of a generally useful image classifier.

Keep the stated limit

Read the labelled picture: Which parameter changed, and which part was not trained? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Keep the stated limit

Change one thing: Change target from 3 to6 and restart v = 0 with the same rate. Predict the result before checking the explanation. What remains the same?

STUDENT LESSON · 7 OF 18

Find and repair a mistake

> Find the unsupported step in the Training a small network claim and explain a repair using the supplied evidence.

Check the claim against the facts

After four updates, freeze v = 1.875. A reserved demonstration pair (3, 6), not used in the updates, receives **prediction** 5.625 and squared error 0.140625. Report it separately from **training** MSE 0.0390625.

One synthetic held-out example is a small demonstration, not **evidence** of broad real-world reliability.

Check the claim against the facts

The frozen representation has a limit: for any negative x, ReLU gives h = 0 and the **output** is always 0.

If the intended task were y = 2x for all real **input**s, x = -1 would need target -2, so this model would fail there regardless of v. **Training** cannot recover distinctions that the fixed hidden representation removes.

Locate the earliest unsupported step

At final v = 1.875, calculate the **prediction** and squared error on reserved pair (3, 6). May its target be included in the four reported **training** updates?

State one repair and the checks you would rerun.

STUDENT LESSON · 8 OF 18

Practise together

Practise Training a small network with support, then explain the deciding step in your own words.

At v = 0, calculate both **prediction**s, errors, MSE and gradient, then update at rate 0.1.

Starting from v = 1.5, calculate both training predictions, errors, MSE, gradient and the next v.

Explain your prediction to a partner before comparing results.

Practise with less help: Check the mean convention

Use workbook W2 for the connected example "**Training** a small network". First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Check your reasoning

Use the worked case for **Training** a small network. Proposed explanation: "Every parameter was trained". Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Answer on your own before discussion. If you are unsure, say which step you need help with.

Explain the picture

Which parameter changed, and which part was not trained? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

My explanation using a pictured step and the given facts

This is supported practice. Keep the separately named independent case for an attempt before feedback.

STUDENT LESSON · 9 OF 18

Run the investigation

> Investigate Training a small network. Record the prediction, actual observation or labelled paper trace, explanation and next test separately.

Predict, investigate and preserve the evidence

At v = 1, the sum of squared errors is 5. A learner uses gradient -10 with rate 0.1. Explain the mismatch with this lesson and give the next v under MSE.

```
train = [(1.0, 2.0), (2.0, 4.0)]
v = 0.0
rate = 0.1
for step in range(5):
    hs = [max(0.0, x) for x, target in train]
    errors = [v * h - target
              for h, (x, target) in zip(hs, train)]
    loss = sum(e * e for e in errors) / len(train)
    print(step, v, loss)
    if step == 4:
        break
    gradient = 2 * sum(e * h for e, h in zip(errors, hs)) / len(train)
    v = v - rate * gradient
```

Record	Your evidence
Given input and fixed rule	
Prediction before checking	
Actual result, or “not run”	

Record	Your evidence
First mismatch and one explanation	
Boundary or missing input and expected response	

On paper, trace the supplied procedure. If code is provided, predict before running it in a local practice file. Never write an expected answer as though it were an observed run.

Plan → try → check

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

STUDENT LESSON · 10 OF 18

Build your understanding

> Move from recall to tracing, diagnosis and transfer in Training a small network. Explain why each step is justified.

Explain → apply → challenge

At v = 0, calculate both **prediction**s, errors, MSE and gradient, then update at rate 0.1.

> At final v = 1.875, calculate the **prediction** and squared error on reserved pair (3, 6). May its target be included in the four reported **training** updates?

For the lesson architecture at x = -1, give h and **prediction** for any finite v. If target is -2, calculate squared error and explain why changing v cannot fix it.

Explain which evidence supports your conclusion and which claim remains unproved.

STUDENT LESSON · 11 OF 18

Connect your project

> Apply this week's idea to Image-pattern Network Investigation. Keep an annotated decision and the evidence supporting it.

Add a labelled **training** a small network **evidence** sheet to Image-pattern Network Investigation. Show how you can trace repeated forward loss gradient and update steps.

Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

Project evidence	Record
Component and version	
Inputs permitted at prediction time	
Trace and comparison with earlier work	
Failure, limit and human review	

Evidence for your project

For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Show what you know

> Use a genuinely new case for independent evidence. Previously discussed examples remain practice; use the separately named independent case before feedback.

Save the model formulas, frozen values, trained parameter, initial value, data pairs, loss definition, learning rate, update budget and the printed trace. State that these numbers are a classroom experiment.

Changing a setting after seeing reserved outcomes makes those outcomes development **evidence** for that revision.

> **A team reports that its training MSE fell each step and concludes the network learned y = 2x for every real x. Give one numerical counterexample and a more accurate conclusion.**

Next week separates an update, a batch and an epoch when **training** uses more than one group of examples.

For the lesson architecture at x = -1, give h and **prediction** for any finite v. If target is -2, calculate squared error and explain why changing v cannot fix it.

My limited claim and a useful next check

STUDENT LESSON · 13 OF 18

Study a complete case

> Revisit the complete Training a small network case for supported practice. Repeated examples do not become fresh assessments.

Complete evidence record

One full-data update from v is 1

STAGE	COMPUTE WITH CURRENT V	RESULT
Forward and loss	Predictions [1, 2] for targets [2, 4]	Errors [-1, -2], MSE 2.5
Gradient	2 times (-1 times 1 - 2 times 2) / 2	Gradient -5
Update with rate 0.1	1 - 0.1 times -5	Next v is 1.5

Each row depends on the preceding calculation. Recompute predictions and loss on the next iteration; do not keep using the first gradient after parameters change.

Follow the **input** through the stated procedure to its result. Cover the result first, predict it, then explain every intermediate step. The text/table accompanies the figure so the example can be read without colour.

My complete trace of the worked case

STUDENT LESSON · 14 OF 18

Try a new case independently

Attempt the reserved new case before feedback. Record support; the earlier examples below remain additional practice.

Independent case: Run a new training case independently

Keep h = ReLU(x), train only v, and use one pair (2, 2), v = 0 and rate 0.125. Use squared error and gradient 2 times (2v - 2) times 2. Perform two updates, reporting each new v and loss.

Explain why the second update changes nothing.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Attempt this without hints

Keep h = ReLU(x), train only v, and use one pair (2, 2), v = 0 and rate 0.125. Use squared error and gradient 2 times (2v - 2) times 2. Perform two updates, reporting each new v and loss.

Explain why the second update changes nothing.

> **My independent procedure, result and limitation**
>
> | |
> |---|
> | |
> | |
> | |
> | |

Record support used: none, hint or teacher explanation. After feedback, a different case is needed to demonstrate independence.

Use feedback to improve

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

Your lesson route

> Use this map to navigate Training a small network, connect the goals to tasks and keep one evidence record for each attempt.

Start with this question: Train a single **output** weight while clearly marking the frozen hidden computation.

By the end, I can	How I will show it
Trace repeated forward loss gradient and update steps	Explain the deciding step, show a trace or test, and state one limit.
Keep frozen parameters distinct from the trained parameter	Explain the deciding step, show a trace or test, and state one limit.
Report training results with their evaluation limits	Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

STUDENT LESSON · 16 OF 18

Finish and return

> Explain Training a small network without notes, identify a remaining question and plan a later check with a changed case.

Close your notes. Explain this goal in your own words: Trace repeated forward loss gradient and update steps

> **My deciding step, evidence and one remaining question**

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

STUDENT LESSON · 17 OF 18

Supported practice, repair and portfolio

> Attempt supported questions before using hints or feedback. Review a missing building block, then keep the completed work.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Train a single **output** weight while clearly marking the frozen hidden computation.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? v becomes1; **prediction** 3 and loss 0. Next gradient is 0.

1. Freeze hidden step

2. Update and verify

3. Compute initial gradient

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Update and verify v becomes1; **prediction** 3 and loss 0. Next gradient is 0. Only the **output** weight v is being fitted in this toy network; the hidden ReLU is fixed.

The **training input** gives hidden value three. Squared-error slope points v from zero towards one, and this specially chosen rate reaches one in a single update.

Zero loss on one **training** pair is exact **fitting** of that pair, not **evidence** about other **input**s.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: Change target from 3 to6 and restart v = 0 with the same rate. Which conclusion is supported by the supplied facts?

1. First gradient=-36; v becomes2; **prediction** 6 and loss 0.

2. We can decide the result without examining the changed **input** or the procedure.

3. This one example guarantees correct results for every future **input**.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: First gradient=-36; v becomes2; **prediction** 6 and loss 0. The hidden value remains3, but the target residual and required **output** weight double.

Repair a missing building block

Review Updating parameters (ai-5/11). Goal: Apply a learning rate to an ordered gradient. Short practice: Why must new loss be recomputed rather than assumed smaller?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Slopes and local change (ai-5/9). Goal: Calculate an average rate of change over a stated interval. Short practice: Which calculation disproves “any increase helps”?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Forward passes by hand (ai-5/6). Goal: Record scores and activations in dependency order. Short practice: Why should feedback identify h before correcting only y?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Predict, run and debug

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

STUDENT LESSON · 18 OF 18

Visual concept journey

> Explain each Training a small network graphic using its labels, deciding evidence and relevant limitation.

01. Training a small network: the case

AI ACADEMY / AI-5/12 · 01 · LOOK AT THE FACTS

Training a small network: the case

The situation

Train a single output weight while clearly marking the frozen hidden computation.

Supplied evidence

Toy h=ReLU(x), y=vh. One training pair x = 3, target 3; start v = 0; squared loss; rate 1/18.
All inputs and outcomes in this worked example are invented classroom data; no real model run is claimed.

By the end, I can explain training a small network using a traced example and a stated limitation.

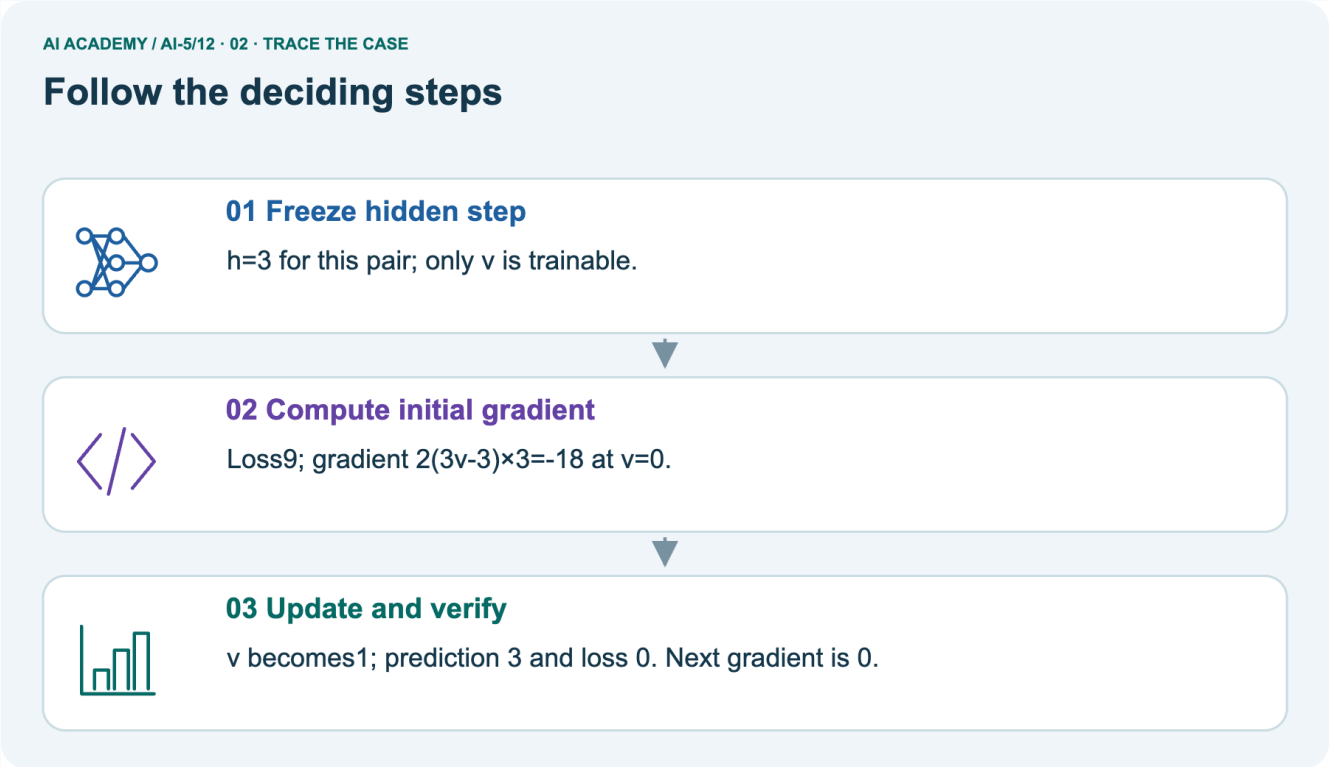
Follow the labelled picture with the matching explanation. Open the vector diagram

Train a single **output** weight while clearly marking the frozen hidden computation. Supplied facts: Toy h=ReLU(x), y=vh. One **training** pair x = 3, target 3; start v = 0; squared loss; rate 1/18.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

> Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

Freeze hidden step: h=3 for this pair; only v is trainable. → Compute initial gradient: Loss9; gradient 2(3v-3)×3=-18 at v=0. → Update and verify: v becomes1; **prediction** 3 and loss 0. Next gradient is 0.

> Point to each numbered stage and explain which recorded **input** it uses.

03. One full-data update from v is 1

AI ACADEMY / AI-5/12 · 03 · INSPECT THE LESSON MODEL

One full-data update from v is 1

Stage	Compute with current v	Result
Forward and loss	Predictions [1, 2] for targets [2, 4]	Errors [-1, -2], MSE 2.5
Gradient	2 times (-1 times 1 - 2 times 2) / 2	Gradient -5
Update with rate 0.1	1 - 0.1 times -5	Next v is 1.5

Each row depends on the preceding calculation. Recompute predictions and loss on the next iteration; do not keep using the first gradient after parameters change.

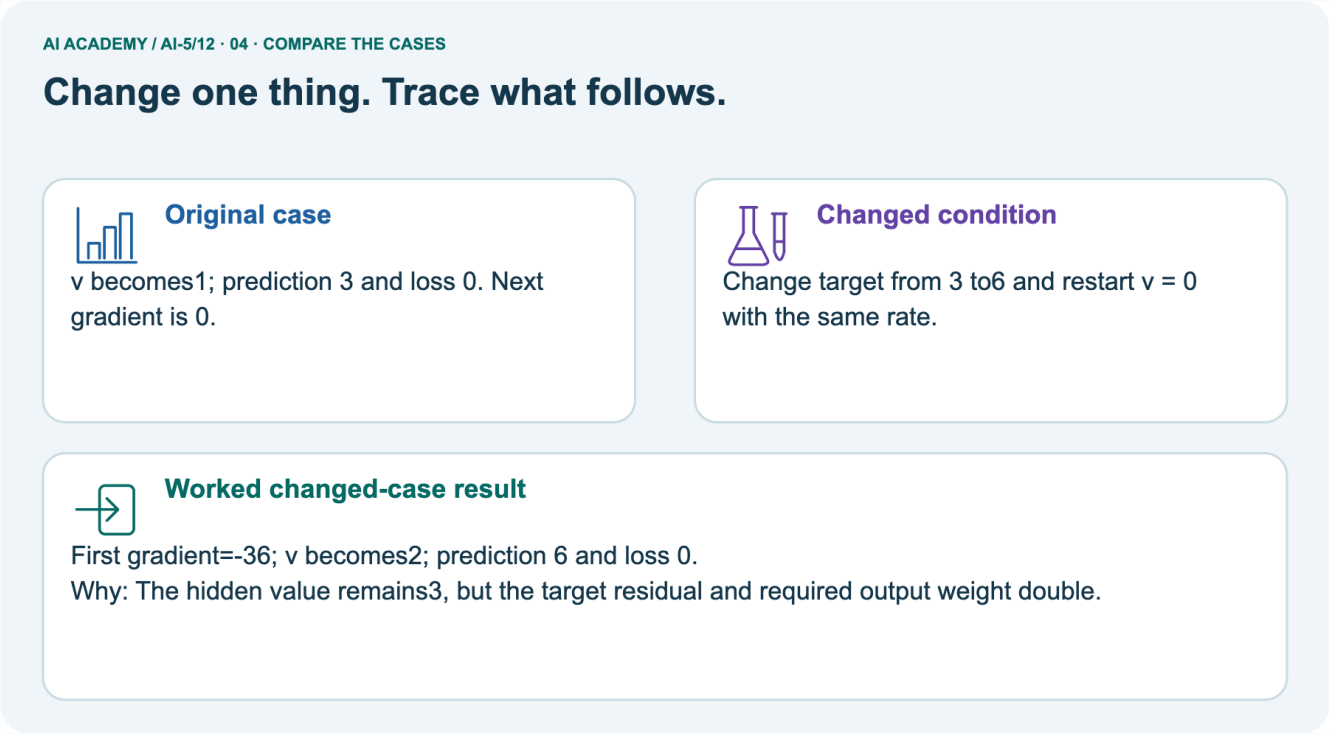
Follow the labelled picture with the matching explanation. Open the vector diagram

One full-data update from v is 1. Forward and loss | **Prediction**s [1, 2] for targets [2, 4] | Errors [-1, -2], MSE 2.5. Gradient | 2 times (-1 times 1 - 2 times 2) / 2 | Gradient -5.

Update with rate 0.1 | 1 - 0.1 times -5 | Next v is 1.5

> Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



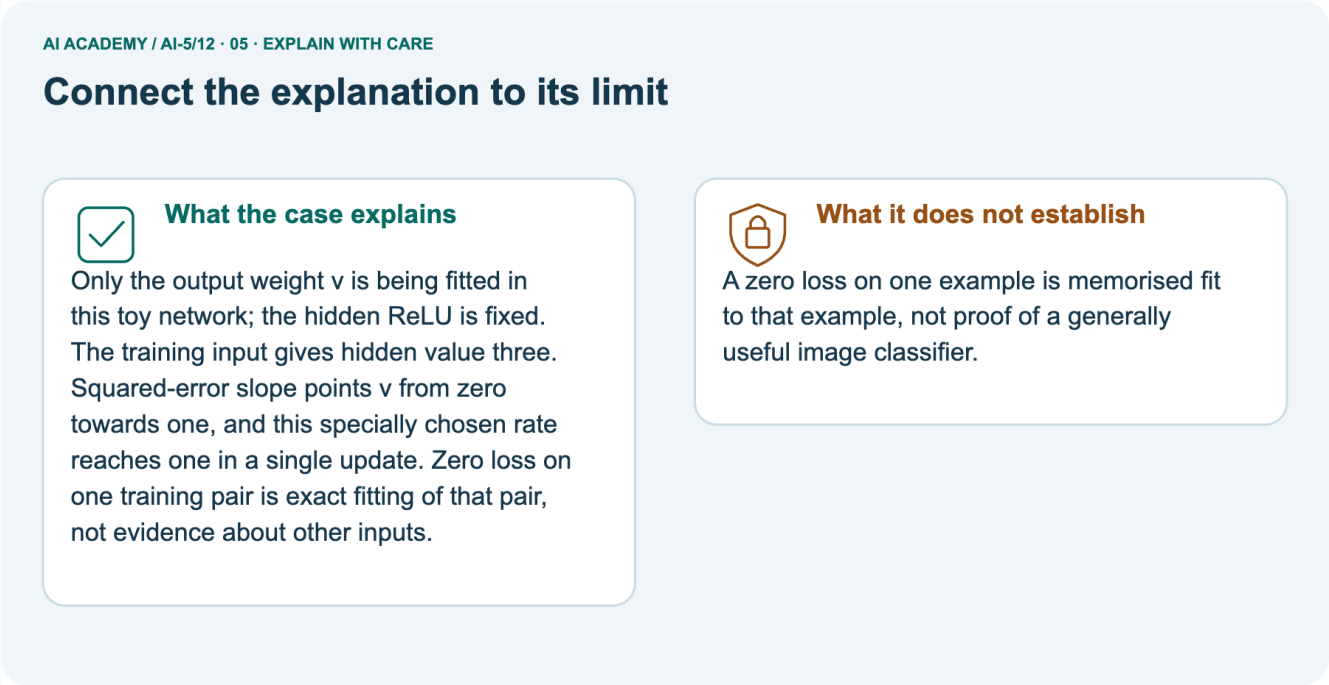
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: v becomes1; **prediction** 3 and loss 0. Next gradient is 0. Changed condition: Change target from 3 to6 and restart v = 0 with the same rate.

Expected result: First gradient=-36; v becomes2; **prediction** 6 and loss 0. Reason: The hidden value remains3, but the target residual and required **output** weight double.

> Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

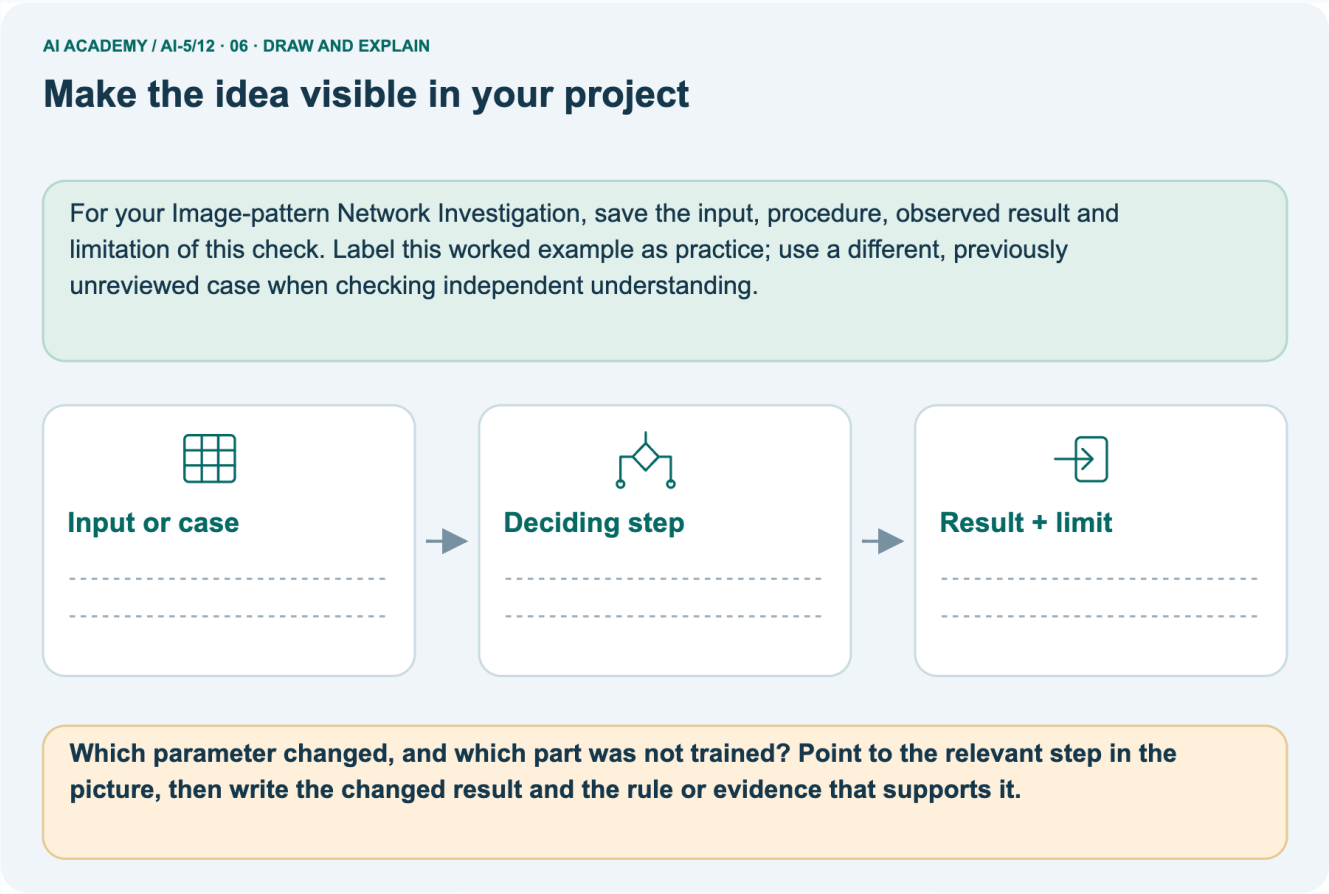
Only the **output** weight v is being fitted in this toy network; the hidden ReLU is fixed. The **training input** gives hidden value three.

Squared-error slope points v from zero towards one, and this specially chosen rate reaches one in a single update. Zero loss on one **training** pair is exact **fitting** of that pair, not **evidence** about other **input**s.

Limit: A zero loss on one example is memorised fit to that example, not proof of a generally useful image classifier.

> Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: Which parameter changed, and which part was not trained? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

> Draw your project trace and justify the deciding step using **evidence**.

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.